

**Question:**  
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

**Reasoning + Answer:**

<think>

The human brain shows marked maturation over childhood and adolescence , with the cere bell um developing much earlier than the cortex . Evidence includes :

- My el ination : The cere bell um is among the most rapidly my el inating regions , with early Pur kin je neuron and gran ule cell synaptic inter connections , whereas cortical circuits ( hip poc ampus , ne oc ortex ) begin my el ination later and are highly interconnected within older childhood .

- Development al trajectories : Vest ib ular and vestib ulo - ocular functions controlled by cere bell ar circuit ry are well established at birth . In contrast , visual ac uity and cortical mat uration depend on ongoing refinement of retinal gang lion cell connections and cortical inputs , leading to a less mature visual system at birth .

- Neuro chemical /biological markers : The cere bell um is enriched in ch oline ac etyl transfer ase and ac etyl ch oline receptors early in life , reflecting early post natal development , whereas cortical regions display higher rates of growth hormone and neuro troph ins over time .

These patterns of early mat uration support the cere bell um as the least developed brain region at birth .

Correct Answer : F . cere bell um

</think>

<answer>

F . cere bell um

</answer>

